

From Imitation Collapse to Visitation-Corrected Distillation: Training a 2B-Parameter MMORPG Tool-Use Agent

A three-round on-policy distillation case study (Kaetram)

Kaetram Agent Project

Technical report, revised July 19, 2026 — single-codebase case study

Abstract

We audit an end-to-end effort to post-train Qwen3.5-2B for long-horizon play in Kaetram, an open-source MMORPG exposed through 17 typed tools. Project notes report three relevant patterns. First, a cross-vocabulary SFT lane fell below the reported base runs while aggregate tool use moved toward the training-corpus marginal. Second, one natural-visitation reverse-KL OPD run changed tool mix and interaction tempo without changing its 12/30 Core-3 score. Third, a later round that added hand-selected persistent player-state starts was followed by a recovery-off 15/30 run and wall passage across three clustered prompt variants; a subsequent weights-plus-recovery system run reported 18/30. These are descriptive observations, not causal effects: there is one unmatched run per cell, prompt variants are not independent seeds, and the June raw bundles, gameplay seeds, checkpoint/configuration digests, and exact render contracts are absent. A historical probe also reports positive teacher-over-student advantage for one malformed continuation, but it does not score both canonical and malformed candidates in the same matched context and therefore does not establish preference reversal. This report preserves the engineering history while explicitly separating what was observed from what remains to be tested: matched fresh-student state-source arms, training-level replication, a weights-by-recovery factorial, state-by-history controls, held-out transfer, reset attestations, and immutable artifacts. The result is an audited case study and confirmatory agenda, not an established method claim.

1 Introduction

Can post-training instill genuinely long-horizon, tool-mediated competence in a $\sim 2\text{B}$ -parameter model — as opposed to surface style? This report traces a single project’s complete answer-so-far across three training paradigms, eleven rounds, and one environment, with every claim tied to logged rollouts and code.

The task: autonomously progress three quest chains (“Core 3”, 10 stages total) in Kaetram, a 2D MMORPG, via structured tool calls only. The metric: new quest stages per 6-hour run, summed over three personality-variant agents (*stages/30*). The arc:

1. **r1–r9 (SFT iteration)**. Nine rounds of supervised fine-tuning of Qwen3.5-9B on Claude-teacher trajectories established methodology (loss masking, prompt parity, truncation gates, a $\sqrt{r}$ LoRA trap) but never beat the base model.
2. **r10 (exploratory negative)**. All four reported base runs exceed all three reported SFT runs. Aggregate decision-verb frequencies are consistent with marginal imitation, but raw artifacts and matched-state probabilities are absent.

3. **r11a (scaffold scouting).** Reported harness and state-contract changes coincide with large score movement across a small, duration-mismatched model-size sweep. The non-monotone sizes motivate a harness hypothesis; they do not establish that capacity is irrelevant.
4. **r11b (this study’s focus): three rounds of on-policy distillation,** scaffolded Qwen3.5-4B (17/30) $\rightarrow$ base Qwen3.5-2B (12/30). Round one transferred the teacher’s *behavioral distribution* while its reported score remains 12/30. A later state-augmented round is followed by a recovery-off 15/30 run and wall passage across three clustered prompts. Round three combines new weights and a recovery affordance and reports 18/30. Missing raw bundles, exact parity, and gameplay seeds prevent causal attribution of the monotone descriptive sequence 12 $\rightarrow$ 12 $\rightarrow$ 15 $\rightarrow$ 18.

Contributions. (i) An evidence audit that separates descriptive behavior movement from unsupported competence and mechanism claims. (ii) A prospective, falsifiable protocol for direct persistent-player-state initialization, with matched reset/prefix baselines and a state-by-history ablation; no method effect is currently claimed. (iii) A bounded tool-interface diagnostic: historical notes report positive teacher-over-student advantage for one malformed continuation, which motivates a paired copy-prior test but does not establish preference reversal. (iv) A weights-by-recovery factorial and immutable-artifact contract designed to prevent the confounding present in the historical 18/30 system result.

2 System

2.1 Environment and benchmark

Kaetram-Open is a tile-based MMORPG: a Node game server per agent (WebSocket), MongoDB persistence, browser client. **Core 3** totals 10 stages: *Foresting* (3 stages; two 10-log deliveries), *Herbalist’s Desperation* (3: accept $\rightarrow$ 3 blueilly $\rightarrow$ 2 paprika + 2 tomato; herbs require Foraging level 5), *Rick’s Roll* (4: talk $\rightarrow$ 5 cooked shrimp $\rightarrow$ a door puzzle $\rightarrow$ delivery). Two mechanics make this hard for an LLM agent: *skill gating* (resources refuse interaction below the required level) and *stochastic yield* — each server tick a harvest succeeds with low probability ($\approx 1/3$ for an unleveld player, rising with skill level), and success depletes the bush. Long-horizon, partially random gathering — not a scripted sequence. The metric, CORE-3, is new stages reached: per agent-run the first-to-last observed stage delta (so resume-state replay cannot inflate it), summed over the three agents, in $[0, 30]$.

2.2 Agent harness

The agent runs an OODA loop (`play_qwen.py`): an OpenAI-compatible client against a Modal SGLang endpoint plus an MCP stdio client driving a Playwright-controlled browser. Decoding uses Qwen3.5’s thinking-mode sampling preset (temp 1.0, top-p 0.95, top-k 20, presence 1.5), but the generation prompt is the template’s *non-thinking* default — a closed-empty `<think></think>` block, so reasoning surfaces as plain content (the dangling-`</think>` dialect, §5.3), a regime held identical across base, teacher, and student. Context budget 14,336 of the 16,384 training length, measured with the same patched chat template the server renders; overflow triggers a *session rollover* (fresh conversation, persistent game state, a programmatic — never model-authored — advisory note carrying quest stage and position). The MCP server exposes 17 tools with three design principles: *enforced affordances* (attack auto-loots; navigate does BFS with snap-to-walkable; craft/buy auto-walk), *advisory state-contract* (`query_quest` leads with canonical facts —

accepted/stage/needed/have/remaining — plus a recommendation the agent must verify; **observe** enriches mobs with level/aggression and quests with item progress), and *payload compaction* (ASCII map dropped; $\sim 2\times$ turns per session). Three agents run in parallel (personality variants: grinder / completionist / explorer — priority modifiers only), each with its own game server, sandbox, and display; `orchestrate.py` supervises; runs are launched from a wiped, freshly-seeded database.

2.3 Training and serving infrastructure

Each model size is a Modal SGLang app (student L4, teacher A100) exposing `/v1/chat/completions` for rollouts and `/v1/score` for teacher-forced per-token logprobs over a caller-supplied (context, full-text) pair — raw pre-rendered strings, because the Qwen3.5 repos ship different chat-template revisions but share one vocabulary, so an identical string tokenizes identically on student and teacher. All components apply one chat-template patch (the stock template silently strips `<think>` from intermediate turns; QwenLM/Qwen3 #1831). Training runs on a Modal H100 with Unsloth + fresh LoRA ($r=64, \alpha=64$, rsLoRA off); Qwen3.5’s 3:1 Gated-DeltaNet hybrid requires pinned fast kernels (`triton==3.3.1 + fla 0.5.0`), which we adopted only after a three-way fp32 parity probe (fast-bf16 vs fallback-bf16 vs fp32 reference on real corpus tokens; the fast path proved marginally *closer* to fp32 than the fallback, grad cosines ≥ 0.997 , $8.9\times$ speedup — round-2 training: 50 minutes vs round-1’s 8 hours).

3 The SFT era (r1–r10): imitation collapse

Rounds r1–r9 (Qwen3.5-9B, Claude-trajectory SFT) produced methodology, not wins: completion-only loss masking (and the discovery that three rounds had silently trained full-token); chat-template CoT preservation; byte-identical train/eval prompt parity; strict 16K truncation gates; and the rsLoRA trap ($\alpha=r$ under rsLoRA is an $8\times$ effective-LR error that diverged r7). r9 still lost to base.

r10 is a motivating negative result whose raw artifact bundle is absent. Project notes describe 8,510 training records from 15 Claude runs. They report base 7/30 in *all four* eval runs (zero variance); r10-SFT scored $\{3, 1, 2\}$, mean 2.0/30 — a $3.5\times$ regression (Mann-Whitney exact, one-sided, $p=0.029$; Fisher’s exact on Foresting completion $p=0.016$, OR= 16). The agent-level Fisher analysis is pseudo-replicated, and neither result is independently regenerable from this clone. Aggregate tool mix is consistent with a hypothesis: the SFT policy’s decision-verb frequencies match the *training-target* distribution within $\sim 1\text{pp}$ (`interact_npc` corpus 2.4% $\leftrightarrow$ SFT 2.1% vs base 10.8%; `query_quest` 2.2 $\leftrightarrow$ 2.5 vs 11.2; `navigate` 27.7 $\leftrightarrow$ 26.4 vs 5.5). Credit analysis showed why that marginal is poison: 52% of source sessions advance no quest; `interact_npc` carries the highest near-term credit (36% of uses precede an advance within 5 turns) yet is 3.8% of the corpus, while `attack` is 25.5% of corpus with $\approx 0\%$ credit. These aggregates motivate a corpus-prior hypothesis rather than a matched-state mechanism result. Possible explanations include compounding off-policy error (DAgger’s $O(\epsilon T^2)$), capacity-gap marginal-fitting, CE diversity collapse, and capability suppression.

4 The scaffold reframe (r11a): harness scouting

Before more training, we asked what a *frozen* model could do with a better model–environment contract. Over eight days the base-9B Core-3 best-run envelope rose through 4, 9–10, 12, 15, and 19/30 while harness/state-contract changes accumulated (quest-list normalization, canonical `current_step` facts, navigation hardening). Durations and configurations differ, so the sequence

Table 1: **Program-wide comparison — quest progression and scale.** Every run in this report, grouped by harness era (the central variable). Quest columns count agents (of 3) completing each chain; Core-3 is new stages reached (last–first observe) summed over the three agents. The R11 block is the controlled comparison (identical scaffold, varying capacity); pre-R11 rows are different-harness reference points (Claude on the Claude Code CLI). Error rate is the game-state failure rate — the high-error rows (Claude, OPD-r3) are attempting the hardest content (Rick’s door; the hostile Herbalist zone), not failing the basics. Durations differ (3–6h) so tool calls/deaths are not duration-normalized; Core-3 (a first-to-last delta) is the duration-robust metric.

Model	Core-3/30	Forest.	Herb.	Rick’s	dur	calls	err%	deaths
<i>pre-R11 harness — different from the R11 Qwen ladder</i>								
Claude Sonnet	27	3/3	3/3	2/3	5.9h	5,605	12.9	114
base-9B (r10-era)	7	2/3	0/3	0/3	3h	1,738	5.1	36
r10-SFT 9B	2	0/3	0/3	0/3	3h	1,610	13.3	62
<i>R11 scaffold — controlled: identical harness, varying capacity</i>								
27B	15	3/3	1/3	0/3	4.2h	2,324	3.4	146
base-9B	19	3/3	3/3	0/3	5h	2,844	3.4	145
4B (teacher)	17	3/3	2/3	0/3	5h	2,140	4.3	105
base-2B (student)	12	3/3	0/3	0/3	6h	6,183	8.6	142
OPD-r3 2B (+recovery)	18	3/3	3/3	0/3	6h	8,563	13.0	183

does not identify each change’s effect. The reported full-run comparison is $7 \rightarrow 19$ (Table 1). It motivates a possible knowing–doing gap at the boundary between advisory and enforced behavior: every affordance that works is *enforced* (auto-loot, BFS navigate); every remaining blocker is *advisory* (survival rules the model quotes and ignores).

The scaffold transfers across capacity. With identical scaffold and prompts: 27B = 15/30 ($n=1$, ~ 4 h), 9B = 12–19/30, 4B = 12/30 (3h) and **17/30** (5h), 2B = 13/30 (5h) and 12/30 (6h). Foresting is solved at every size and Rick’s Roll is essentially untouched at every size (only the 9B’s best run reaches Rick’s stage 1); the *Herbalist wall* is where capacity shows — the 2B stalls at stage 1 while 4B, 9B, and 27B all clear it (the 27B also gained a tool-result confabulation failure mode). Above the 2B, the observed scores are not monotone in capacity — the 27B’s 15 sits inside the 9B’s 12–19 — but durations and n differ. The reported 4B 17 vs 2B 12 gap motivates the distillation experiment: **teacher = scaffolded 4B, student = base 2B** — a real, same-family competence gap, an order of magnitude cheaper per round than the 9B lane. (Reference ceiling, on the pre-R11 Claude Code harness: in its best run 2/3 of the Claude agents complete *all* of Core 3 including Rick’s Roll — 27/30, Table 1 — in 6h at 1,400–2,300 tool calls. This shows the game is winnable, not a same-harness ceiling for the Qwen ladder.)

Program-wide comparison. Tables 1 (quest progression & scale) and 2 (tool-use signature) collect every run in this report, grouped by harness era. They make three things visible at once: the *harness lift* (the same base-9B goes $7 \rightarrow 19$ from the pre-R11 to the R11 harness), the *SFT regression* (r10-era base 7 $\rightarrow$ r10-SFT 2, with the tell-tale navigate amplification and decision-verb suppression in Table 2), and the *capacity ladder* at fixed R11 harness (2B 12 / 4B 17 / 9B 19 / 27B 15, saturating above 9B) into which OPD lifts the 2B ($12 \rightarrow 18$). Only the R11 block is a controlled comparison; the pre-R11 rows (Claude, r10) are different-harness reference points.

Table 2: **Program-wide comparison — tool-use signature** (% of tool calls; same rows and grouping as Table 1). The r10-SFT collapse is legible: `navigate` amplified to 27.9% (from base-9B’s 5.2%) while the decision verbs (`query_quest`, `interact_npc`) are suppressed — the corpus-marginal imprint. The R11 models lean on `query_quest` and `gather` (the scaffold’s progression verbs).[†]Claude’s turns/session is `maxSessionTurns`-bounded, not comparable to the Qwen agents’ 16K-context rollover sessions; use *tool calls* (Table 1) as the cross-harness scale.

Model	observe	navig.	attack	gather	q_quest	npc	turns/s
<i>pre-R11 harness</i>							
Claude Sonnet	28.2	23.6	18.7	11.3	1.5	3.2	151 [†]
base-9B (r10-era)	39.5	5.2	15.5	4.3	4.1	8.7	3.7
r10-SFT 9B	32.7	27.9	13.9	5.5	2.1	2.7	3.7
<i>R11 scaffold</i>							
27B	31.9	22.0	2.8	17.7	8.8	1.5	8.2
base-9B	36.5	13.6	10.0	12.9	11.4	2.1	6.1
4B (teacher)	36.1	9.1	9.1	20.8	8.7	1.4	6.1
base-2B	26.6	14.3	17.9	17.9	5.3	3.8	9.1
OPD-r3 2B	30.3	21.2	10.1	11.4	13.3	3.1	8.0

5 On-policy distillation: method

5.1 Objective and loss

Let π_S , π_T be student and teacher, and d^{π_S} the state distribution induced by student rollouts (a “state” is the full serving prompt: system, bootstrap, dialogue history). Batched reverse-KL OPD minimizes

$$\mathcal{L}_{\text{OPD}} = \mathbb{E}_{s \sim d^{\pi_S}} \mathbb{E}_{a \sim \pi_S(\cdot|s)} \left[\log \pi_S(a | s) - \log \pi_T(a | s) \right], \quad (1)$$

the I-projection (mode-seeking) direction, estimated on the student’s own emitted tokens with the teacher as a per-token grader. We freeze a per-token advantage at data-build time — the negated student-minus-teacher logprob gap, taken against the *behavior* (rollout-time) logprobs — and train with PPO-style clipped importance sampling ($\epsilon=0.3$, advantage clamp ± 3 , plus a $1.5\times$ weight on each session’s first third). The clip is load-bearing: unclipped REINFORCE on a fixed batch’s majority-negative advantages pays the optimizer to drive those logprobs to $-\infty$ and degenerates the policy to an `observe` loop — a collapse we hit and the clip prevents. Because each round’s init equals the rollout-generating policy (fresh LoRA over the previous round’s merged checkpoint), the importance ratio is ≈ 1 at step 0. The clipped objective discourages large local ratios but does not hard-bound probability movement under shared-parameter updates; a step-0 calibration probe verifies the serving-vs-training numeric gap (round 2: median $|\Delta \log p| = 0.0005$, clip fraction 0.000). LR 5×10^{-5} , one epoch, effective batch 32, H100; the LM head is applied only at action positions (a full-vocab forward at $16\text{K} \times 248\text{K}$ would OOM).

5.2 Visitation coupling and environment seeding

Equation 1’s expectation over d^{π_S} is the binding constraint this study makes concrete: the teacher is only ever *queried* at states the student visits. If the teacher’s competence edge lives at states of measure ≈ 0 under d^{π_S} — e.g. standing at a Blue Lily Bush with Foraging ≥ 5 — no amount of dense grading transfers it. Round 1 demonstrates the failure; rounds 2–3 demonstrate the repair: change the *state distribution*, not the loss. Concretely we mix the student’s natural visitation with

rollouts of π_S launched from DB-seeded milestone states — player rows written directly into the game’s MongoDB at e2e-verified mid-quest configurations (quest stage, skills, inventory, position), $\sim 30\%$ of records by count. In the proposed matched experiment only the state source changes. The historical rounds do not meet that condition: checkpoint initialization, collected policy, record count, and data batch also differ. Resetting to hard or intermediate states to get learning signal there is a well-trodden idea in RL — reset along a demonstration [19], Backplay’s reverse curriculum [20], Go-Explore’s restore-then-explore [21], and expert-state reset mixtures for LLM agents [22]; the LLM-side analogues operate in text-prefix or policy space (branched expert anchors [18], behavior-policy blending [24]). Our narrow candidate contribution is a reverse-curriculum state reset instantiated as a *write of persistent, typed environment state* (a DB row), used for on-policy *distillation*. TCOD, Guided-OPD, ReOPD, SCoRe, Backplay, and TRB make a broad state-curriculum claim unavailable; only a matched direct-player-state result could support the narrower distinction.

5.3 Data construction: the prefix property

Each action turn is rebuilt from rollout logs as the byte-exact serving prompt (system + bootstrap + last 28 messages rendered through the patched template with `add_generation_prompt`) plus the **verbatim logged emission** appended directly: full = ctx || emission. Re-rendering the completed turn through the chat template is wrong in a subtle way we hit twice: the template extracts reasoning from any `</think>` inside content and re-renders a real think block, diverging from the closed-empty-think generation prompt the model was actually served — 4% of base-2B turns in round 1, but 99% of r1-policy turns (whose acquired dialect ends content with a dangling `</think>`). Verbatim construction makes the prefix property hold by construction; the round-2 build of 7,849 states had *zero* alignment failures of any kind. Both endpoints score the same raw string; behavior logprobs are therefore exactly the rollout policy’s serving distribution. Every 10th session is held out for the gate.

5.4 Abstention masking and the gate

Advantages are zeroed on token spans matching malformed parameter keys (`<parameter=K=V>`) where the teacher’s grading is probe-verified unreliable (§6.2); 177 spans / 1,422 tokens (0.06%) in round 2. (Round 3 grades these same spans correctively instead of masking them, §8.) The post-train gate scores the new student on the held-out set and reports the token-weighted reverse-KL to the teacher over scorable action tokens (masked spans excluded). That number is *directional only*: round 1 used a $\geq 30\%$ reduction as a PASS bar and it *failed* (+10.9%) while behavior moved dramatically — KAT [7] shows per-token KL on teacher-agreement-trapped spans is a misleading quality signal — so rounds 2–3 gate hard on behavioral checks instead: sampled-completion malformed-rate not worse than the prior round, no degeneration, and no catastrophic reverse-KL blow-up. Round-2 gate +14.5%, round-3 gate +6.4% (diminishing, as expected once the student is near the teacher), both PASS.

6 Round 1: style transfers, competence does not

Setup. Corpus: the 2B’s own 6h baseline rollouts (5,564 train / 574 held-out records) re-scored by the 4B; the pre-train diagnostic put the largest student–teacher disagreement exactly on the 2B’s documented weak verbs (eat.food +0.298 nats, respawn +0.292). Training: 174 steps. Eval: 6h, 3 agents, protocol-identical to baseline (compact observes, fresh DB, parity verified).

Result: Core-3 12/30, identical to baseline — every agent 4/10 with the same split, stuck at the same Herbalist stage-1 wall. Underneath, the student moved decisively onto the teacher’s distribution:

	base-2B	2b-opd-r1	4B teacher
turns/session	9.1	6.1	6.1 (exact)
observe share	26.6%	35.5%	36.1%
navigate share	14.3%	9.4%	9.1%
attack share	17.9%	6.6%	9.1% (overshot)
query_quest share	5.3%	18.0%	8.7% (overshot 2×)
chars/turn	411	1248	864 (overshot)
game-state error rate	8.5%	5.5%	4.3%

Execution improved where the signal pointed (navigate errors 5.3% $\rightarrow$ 1.2%; eat-inedible halved; phantom-target re-attacks replaced by re-grounding observes), with textbook mode-seeking signatures: exact convergence on the teacher’s high-frequency modes and amplification *past* the teacher on its preferences.

6.1 The KL×behavior cross-table

The held-out gate moved only +10.9% (a formal FAIL against its 30% bar) while behavior moved dramatically — and the per-verb *signs* line up: every verb whose rKL dropped (attack −17%, stuck_reset −23%, warp, respawn, navigate) moved toward the teacher behaviorally, and **query_quest** — the single verb whose rKL *increased* (+20%) — is exactly the verb that behaviorally overshot the teacher 2×: the student moved *past* the teacher and diverged from the other side. Small per-token movement compounds over autoregressive rollouts; a token-KL gate on old-visitation states measures the wrong quantity.

What the advantages reward: structure, not reasoning. Reading the round-1 held-out advantages directly (teacher minus student logprob over the emitted tokens), the positive reverse-KL signal concentrates on tool-call *structural* tokens — the `</think>` delimiter (+0.212, 79% positive) and the `<function=</parameter=>` syntax (sharply positive) — while *reasoning-prose* tokens carry the most *negative* mean advantage (−0.333; the tool-call region as a whole averages $\approx$ 0). The `</think>` reward — the dialect’s entry hook — explains the tag rate jumping 4% $\rightarrow$ 99%; the $\sim$ 4× growth in reasoning *length* is instead a trajectory-level mode-seeking pull toward the teacher’s reason-then-act trajectories, *not* a per-token reward. So OPD transfers the *shape* of reasoning, not its substance — and the round-3 compression (§8) is the same signal with that pull relaxed.

6.2 The teacher-forcing copy prior

Round 1 introduced a defect: 8.3% of argument-requiring calls arrived with empty input, dominated by one span family — the model writes the boolean kwarg into the parameter *key*, `<parameter=accept_quest_offer=True>` (154 malformed vs 18 correct in the r1 eval; 0 vs 182 in baseline), which the parser drops. Historical notes report teacher log probability −0.161 for continuing that malformed candidate after malformed history, versus −0.253 under the student. This is positive teacher-over-student distillation advantage for one candidate, not evidence that the teacher prefers it over the canonical candidate. The reported correct-form observation uses a different state, and the raw probe artifact is absent. A paired clean/malformed-history by

canonical/malformed-candidate experiment is therefore required. CCOPD, SOD, KAT, and per-context OPD diagnostics also preempt a broad degraded-context novelty claim; the remaining copy-prior framing is a hypothesis.

7 Round 2: visitation-corrected distillation

Environment-state seeding. The 4B teacher passes the Herbalist wall in its own runs (2/3 agents finished the quest) — the gap is real and teacher-demonstrated — but base-2B rollouts essentially never visit lily-bush-at-Foraging ≥ 5 states (20 gated attempts in 6,183 calls). We therefore seeded the *environment database*: player rows written mid-quest (Herbalist stage 1 accepted, Foraging XP 450 = level 5 on the game’s XP curve, starter kit, spawn at the e2e-verified walkable tile beside the lily cluster), then a 2.5h collection run of the r1 policy from those states. The intended control point is $d^{\pi s}$, but the historical round differs in more than state source. In the seeded rollouts all three agents passed the wall (the first 2B passages ever observed), producing both student-authored success trajectories and dense teacher grades at decision states. This is a reverse-curriculum state reset — the RL lineage (reset-along-a-demo [19], Backplay [20], Go-Explore [21], expert-state resets [22]) and the LLM-side text-prefix/policy analogues (branched rollouts [18], behavior blending [24]); our instantiation, a write of *persistent typed environment state* for on-policy *distillation* (not reward-RL), is what we have not seen published.

Corpus and training. 7,849 states (5,523 natural r1-eval rollouts + 2,326 seeded wall states, interleaved) $\rightarrow$ 7,024 train / 825 held-out records, zero construction failures; fresh-state per-verb rKL came in at $\approx$ half of round-1 levels (eat_food +0.298 $\rightarrow$ +0.149; Herbalist frontier +0.261 $\rightarrow$ +0.136) — round 1 had genuinely moved the policy toward the teacher on its own visitation. 220 steps, 50 minutes. Gate: **PASS** — held-out rKL 0.4281 $\rightarrow$ 0.3662 (+14.5%, *every* verb improved, including query_quest reversing its round-1 sign), no degeneration, no blow-up.

Eval (6h, unseeded, parity verified): Core-3 15/30 — 5/10 for every agent. All three agents passed the Herbalist stage-1 wall unseeded (at 4.1h, 4.5h, 5.0h) — a stage neither base nor r1 ever reached: arm-level passage rates 0/3, 0/3, 3/3. Supporting movement: all accepts landed by 3.2h (prior arms’ stragglers: 5.7h, 5.3h); correct-form accepts 26 vs 86 malformed (r1: 18 vs 154); Foresting completion monotone faster across rounds (0.4–0.6h vs 0.5–0.8 vs 0.7–1.8); the round-1 overshoots retreated toward the teacher (query_quest 18.0 $\rightarrow$ 12.9%; verbosity 1248 $\rightarrow$ 1116 chars/turn), with turns/session holding at the teacher’s tempo (6.1 $\rightarrow$ 6.0). Qualitatively, the passage mechanism is visible in the logs: r2 agents rotate bushes after empty harvests, track the Foraging-5 gate explicitly, and exploit gather’s auto-walk; r1 at the same wall exhibited a *premature-turn-in attractor* (standing one tile from the bush holding 2/3 lilies, it walks 85 tiles to the quest NPC with an incomplete set) and gate-blind retries at Foraging 1–4; base displaced into combat (a level-59 character with Foraging 4). At run end all three r2 agents were mid-stage-2 (one paprika + tomatoes held; the completionist made the first-ever deliberate trip toward the paprika cluster and nav-looped six tiles short).

Costs. (1) The malformed-call attractor *mutated rather than died*: kwarg-in-key declined (154 $\rightarrow$ 86) but a new dominant form appeared — Python-call syntax inside the function tag, `<function=gather("Oak")>`: 0 (base) $\rightarrow$ 79 (r1) $\rightarrow$ **599** (r2) — plus corrupted closing tags. Abstention masking removes reinforcement but supplies no corrective pressure; the model’s own malformed history compounds in-context. (2) The OPD models carry a higher total failure rate than base (base 8.6% / r1 16.9% / r2 16.7%), driven by a validation-failure class ($\approx$ 0.1% at base $\rightarrow$ $\approx$ 11%) that the malformed calls produce; the game-state error class, by contrast, improved across rounds (8.5/5.5/5.9%). (3) Deaths rose monotonically (142/172/188), though 78% of r2

death positions are on the quest NPC’s tile inside a level-45–54 aggro zone: substantially the *cost of attempting the quest* (turn-ins survive death; agents name the danger and proceed).

8 Round 3: counterfactual grading, full-ladder seeding, and a harness affordance

Round 3 targeted the two open wounds — the mutated malformed-call attractor and the stalled Herbalist stage 2 — with two weights-side moves and, ultimately, a harness one. **Counterfactual-canonicalized grading:** at parser-flagged records the teacher scores the student’s emission under a *canonicalized* context s' — the system prompt’s Python-style tool-doc literals rewritten to canonical wire syntax — while the student/behavior side keeps the real context s . Scoring the teacher under s' turns its clean-convention preference into a corrective negative advantage on the malformed tokens, rather than round-1’s +0.09 copy-prior endorsement. This is a designed reward, not an unbiased reverse-KL estimate (student and teacher are scored under different contexts); we adopt it deliberately at exactly the spans where the visitation context poisons the grade. A pre-build *flip probe* set the scope empirically: canonicalizing the malformed *history* calls did nothing (a measured 0% flip), but rewriting the system prompt’s Python-style tool-doc literals in the teacher’s grading copy flipped 86% of flagged states at -1.21 nats — so s' rewrites the doc literals only. **Full-ladder seeding:** two seeded collections placed the 2B at every unsolved stage (Herbalist stage 2; Rick’s fishing/cook/turn-in/door), all graded by the plain 4B. Build: 8,856 train / 1,040 held-out, 4 score failures; 365 counterfactually-graded records (zero masked — corrective grading replaced abstention). Gate PASS (+6.4% rKL, every verb).

The decisive negative result. The first eval was stopped at 35 minutes: counterfactual grading did *not* suppress the malformed *emission* — it regressed it, paralyzing two of three agents (57–76% of their actions were malformed calls the server drops, which fall to text, never execute, and the model re-emits in a spam loop). Historical notes say the grading-context intervention did not suppress student emissions in this stopped run. A *harness affordance* then supplied a runtime workaround: when the server drops a malformed call, the harness recovers the executable call (99.5% coverage), rewrites history to a clean canonical turn (severing the in-context copy prior at its source), executes it, and returns a loud format-correction note.

Reported result: 18/30 for the weights-plus-recovery system, 6/10 per prompt variant. Every agent now completes two Core-3 chains unseeded — Foresting *and* Herbalist — where base completes one (Foresting, then stalls at the Herbalist wall). All three completed Herbalist 3/3 unseeded (round 2 had zero), breaking the stage-2 wall via three concrete differences from r2’s stuck behavior: navigating to the *actual* paprika bush (deep in the hostile zone) rather than the prompt’s stale coordinate; recovering from a one-tile navigation stall by re-targeting an adjacent tile (where r2 looped identically ~ 16 times); and a tighter observe-act loop (half-length reasoning re-observes often enough to notice a frozen position) — with gathered items surviving death. The harness recovery worked and historical notes describe as repeated-marker-free: 405 rewritten sessions are reported to contain one format note and no later marker. The source logs and raw pre-rewrite emissions are not packaged, so this cannot establish model correction or relapse prevention. The same notes report validation/format errors halving (10.7% $\rightarrow$ 5.1%); malformed emissions numbered 335 versus r2’s reported 685. Reasoning compressed sharply against round 2: non-empty median 243 chars vs r2’s 558, $\sim 30\%$ near-empty turns (r2: 0%), revision-churn down (59% $\rightarrow$ 30%), and the dangling-`</think>` dialect abandoned (99% $\rightarrow$ 12%), with grounding preserved.

The historical runs do not identify a causal decomposition. The notes report base 12/30, round two at 15/30 without recovery, round-two weights with recovery at 17/30, and round-three

weights with recovery at 18/30. Each cell contains one unmatched run; raw bundles, exact checkpoint and render parity, gameplay seeds, and immutable manifests are absent. Score subtraction therefore cannot identify weights, recovery, or their interaction. The recovery-off round-two wall passage across three clustered prompts is the strongest historical observation, but it is neither three independent replications nor a pure-weights estimate. The reported throughput contrast ($\sim 40\%$ more turns per hour) is likewise descriptive until reproduced under matched runs.

Rick’s Roll stayed 0/4 — a three-link execution failure, not missing intent. The agents understood and attempted the full chain (52 navigations to Rick, 5 NPC-interaction attempts, 12 door attempts, all quoting the walkthrough). The chain dies at link one: the fishing pole is never re-equipped (Forester’s logs displaced it), so every shrimp gather returns nothing and cook/door/Rick never become reachable; `craft_item` was called *zero* times in all post-Herbalist sessions — exactly the cook-incompetent-teacher prediction (the 4B never cooks, so its seeded-state grades there were weak). Historical notes therefore motivate three separate hypotheses: visitation can expose teacher competence, recovery can address an interface failure, and an incompetent teacher cannot supply useful local supervision. The 18/30 arm is a weights-plus-recovery configuration, not a test that identifies any of those effects.

9 What is and is not valid

Supported observations. (a) The r10 SFT arm fell below base in every one of seven exploratory runs (base [7, 7, 7, 7], SFT [3, 1, 2]; unadjusted exact tests $p=0.029 / 0.016$). The aggregate tool-frequency match is a mechanism hypothesis, not a matched-state causal result. (b) In a small, duration-mismatched capacity sweep, larger models did not monotonically improve Core-3 score; this is scouting evidence, not a scaling conclusion. (c) Round-1 style-without-competence transfer (three independent measurement families agree: tool mix, tempo, per-verb KL signs). (d) The copy-prior mechanism hypothesis (an unpaired historical probe; alternatives remain open). (e) The round-2 lift as a descriptive observation across clustered prompts, not a causal estimate. (f) The round-3 results: the Herbalist stage-2 break and its three-part mechanism (correct coordinate, stall-recovery, tight loop — all read directly from the transition sessions); the reported harness-recovery marker count (405 rewritten sessions, with source logs not packaged); the grader-vs-generation gap (the flip probe verified -1.21 nats of endorsement suppression, yet the stopped run shows the emission *regressed* — a reported falsifying observation); and the reported cook null (`craft_item` = 0, pole-equipped = 0 across all agents).

Faults and caveats, enumerated. (1) *The 18/30 arm is not pure-weights.* It ran with the harness tool-recovery affordance on; base/r1/r2 did not. The unmatched round-2-weights-plus-recovery run reports 17/30, but raw bundles, exact configuration parity, and gameplay seeds are absent. It cannot decompose the round-two-to-round-three difference. Round two’s 15/30 wall passage is recovery-off and the least-confounded historical observation, not a pure-weights effect. (2) *$n=1$ per arm, agents are prompt variants not independent seeds; direction-consistency is suggestive, not significant.* Stage counts are coarse (± 2 –3/30 historical band); error-rate and milestone analyses carry the most descriptive detail but no population inference. The r10 negative is the only historical comparison with multiple reported runs, but its raw bundle is absent; the beyond-base claim — even at 3/3 vs 0/3 on the wall — rests on prompt-variant arms that share a base policy and so are a partial (not independent) replicate, which is why we lean on the categorical wall passage rather than the stage delta. The obvious next validation is seed replication of the round-two passage, which we have not run. (3) *The malformed-emission counter is a lower bound.* The harness rewrites assistant content to canonical before logging, so the counter catches only malformed text that

survived recovery (historical notes report 335 logged vs 405 recovery markers); the true round-3 malformation count sits between the two. (4) *The mixed-context advantage is a designed reward, not an unbiased KL estimate*: the teacher is scored under the canonicalized context while the student is scored under the real one. CCOPD [33] uses the same move without a bias analysis, so we inherit that gap. (5) *Residual analyzer caveats*: quest-transition verb attribution reports the in-flight tool not the cause; **gather** item counts race the server and under-report successes; the `eat_food` error class conflates full-HP/slot-shift with validation empties. (6) *Costs are billing-verified* (**modal billing report**, as for r10’s \$360): r1–r3 totaled \$160 incl. all debugging (~44% of one r10 SFT cycle), but the per-round split is the opposite of intuitive — round 1 ≈\$111 (dominated by a 3× broken-kernel ~8h H100 retrain at ~168s/step), round 2 ≈\$33, round 3 ≈\$16. Once the fast GDN kernels were fixed (round 2) the marginal build+train of a round is ~\$5–10; eval serving on L4 (~\$6–23/round) then dominates. No paid-API cost enters the loop — student rollouts and the 4B teacher are both self-hosted Qwen on Modal. (7) *The decode regime is non-thinking, by inheritance not design*. Every model ran the Qwen3.5 template’s default closed-empty `<think></think>` prompt (the thinking-mode *sampling* preset sits on top of it), so reasoning surfaces as plain content; we did not deliberately choose thinking vs. non-thinking. It is held identical across base, teacher serve, teacher grading, and every student eval, so it is an internal-consistency property, not a confound on any comparison — whether thinking-on would change the results is untested.

Not claimed. No general statement about 2B capacity ceilings (Rick’s failure is multi-causal — a displaced tool, a budget limit, and a teacher null — not a clean capacity result). No claim that grading-context fixes cure generation defects (the stopped run falsifies it). No claim that masking cures teacher-shared defects (the attractor mutated 154 → 86 on one form while a new form went 0 → 599). No cross-environment generality; findings are conditional on a deliberately strong scaffold whose transfer is untested.

10 Related work

Game-agent systems. Orak uses MCP as a plug-and-play interface across twelve games and includes a gameplay fine-tuning dataset [2]; MCP and structured game actions are therefore infrastructure here, not a novelty claim. BALROG evaluates long-horizon language and vision-language agents across games and documents a knowing–doing gap [3], while lngame-Bench studies modular gaming harnesses and transfer from single-game training to unseen games and external planning tasks [4]. Unlike these benchmark-centered systems, our question is how a small policy’s persistent environment-state visitation controls what dense teacher grading can transfer.

On-policy distillation. Reverse-KL on-policy distillation for LLMs was established by GKD [1] and popularized as a post-training recipe by Thinking Machines [5]; the 2026 wave maps its failure modes: trust-region restriction of unreliable teacher tokens [8], token-subset selection [10, 11], structural-token masking [12, 13], verifier-gated sign-consistency [14], trajectory-filter-then-soft-reweight [15], and the KL agreement trap, where low per-token KL on degraded prefixes is the failure signature [7]; DistIL proves reverse-KL-family objectives can raise probability on worse actions [9]. Our historical copy-prior probe does not yet establish a wrong-signed preference; that requires the paired candidate study. The visitation-coupling lens is formalized in States-not-Tokens [17], and our round-1 outcome — style-level convergence without capability gain — is its documented failure mode when the teacher’s edge lies outside the student’s visited states [27]; state-distribution control by resetting to hard/intermediate states is well-trodden in RL (reset-along-a-demo [19], Backplay [20], Go-Explore [21]) and appears LLM-side as branched rollouts from expert anchors [18], expert-state reset mixtures [22], resume-from-verified-prefix [23], and behavior-policy blending [24].

Our prospective distinction is persistent typed player state independent of a successful prefix. Multi-turn agentic OPD at small scale documents per-turn KL escalation and curricula [25]; iterated rounds are the DAgger loop [26, 6]. The harness-scouting hypothesis aligns with the 2026 system-scaling literature (interface adaptation over weights) [30, 31]. The r10 observation connects to imitation collapse / capability-suppression results in SFT [32].

Grading-context engineering (round 3). Scoring student tokens under a teacher context that differs from the generative context is an emerging family: CCOPD grades under a *canonical* clean context to fight self-anchored drift [33]; OPCD conditions the teacher on internalized context [34]; MixSD switches grading context per token [35]; CoD combines original and counterfactually-perturbed teacher contexts off-policy [36]. Our canonicalization of the teacher’s grading context (the system-prompt tool-doc literals) is a historical intervention whose effect on generation is not established. CCOPD preempts broad clean-context novelty; a matched tool-candidate probe is required for any narrower claim. **Privileged-information distillation** is a relevant route for Rick’s Roll, where the same-family teacher is incompetent: a PI-conditioned teacher (verified or frontier-model trajectories in its context) scores the student’s rollouts token-by-token [37, 38], with evidence-masked updates [39]; this enters as prompt text and so sidesteps the cross-vocabulary problem that sank our SFT line, and logit-free chunk-similarity distillation extends it to black-box teachers [40]. On the negative-gradient question, AOPD finds bare unlikelihood reinforcement ineffective and prefers corrective distribution-matching [41] (supporting our corrective-grade-over-mask choice), while AntiSD demonstrates a sign-reversed self-distillation term under an entropy gate [42]. The copy-prior mechanism’s substrate is in-context copy bias and induction-head dynamics [28, 29].

11 Discussion and future work

The historical sequence sharpens the open agenda but does not confirm a mechanism. Full-ladder seeding, new weights, and recovery coincide in the later rounds; their contributions remain unidentified. Rick’s 0/4 is likewise multi-causal: a displaced fishing tool gates link one, the traverse exceeds a session budget, and cooking is never reached. The next useful work is therefore controlled rather than another narrative round: (i) train fresh matched students across natural OPD, direct player-state resets, generic resets, TCOd-style prefixes, and faithful Guided-OPD; (ii) cross weights with recovery while preserving raw emissions; (iii) test privileged context at matched cost and on a held-out quest; and (iv) score canonical and malformed candidates under paired histories. The grading-side probe is a hypothesis about generation, not a guarantee.

12 Conclusion

The historical record motivates three hypotheses rather than establishing three effects. First, the reported natural-OPD round changes tool use and tempo while its score remains 12/30. Second, a later state-augmented, recovery-off round is followed by 15/30 and wall passage across three clustered prompts. Third, a weights-plus-recovery run reports 18/30, while Rick’s Roll remains 0/4. One unmatched run per cell, missing June raw bundles, absent gameplay seeds, and incomplete configuration provenance prevent causal attribution or population inference. The durable result of this report is therefore an audited failure analysis and a matched confirmatory design. Fresh training replicates, reset attestations, a weights-by-recovery factorial, prefix/reset baselines, held-out quests, and paired copy-prior artifacts must decide whether the proposed mechanisms survive.

References

- [1] Agarwal et al. *On-Policy Distillation of Language Models: Learning from Self-Generated Mistakes*. ICLR, 2024.
- [2] Park et al. *Orak: A Foundational Benchmark for Training and Evaluating LLM Agents on Diverse Video Games*. ICLR, 2026.
- [3] Paglieri et al. *BALROG: Benchmarking Agentic LLM and VLM Reasoning on Games*. ICLR, 2025.
- [4] Hu et al. *Imgame-Bench: How Good are LLMs at Playing Games?*. ICLR, 2026.
- [5] Lu and Thinking Machines Lab. *On-Policy Distillation*. Connectionism, 2025.
- [6] Song and Zheng. *A Survey of On-Policy Distillation for Large Language Models*. arXiv:2604.00626, 2026.
- [7] Xin et al. *Escaping the KL Agreement Trap in On-Policy Distillation*. arXiv:2606.09471, 2026.
- [8] Xing et al. *Trust Region On-Policy Distillation*. arXiv:2606.01249, 2026.
- [9] Agrawal et al. *Reinforcement Learning from Rich Feedback with Distributional DAgger*. arXiv:2606.05152, 2026.
- [10] Xu et al. *TIP: Token Importance in On-Policy Distillation*. arXiv:2604.14084, 2026.
- [11] Wang et al. *Not All Disagreement Is Learnable: Token Teachability in On-Policy Distillation*. arXiv:2605.26844, 2026.
- [12] Fu et al. *Revisiting On-Policy Distillation: Empirical Failure Modes and Simple Fixes*. arXiv:2603.25562, 2026.
- [13] Jiang et al. *Cornerstones or Stumbling Blocks? Deciphering the Rock Tokens in On-Policy Distillation*. arXiv:2605.09253, 2026.
- [14] Xu et al. *SG-OPD: Sign-Gated On-Policy Distillation via Sign-Consistency Gating and Phased Teacher Sampling*. arXiv:2606.09304, 2026.
- [15] Li et al. *Filter, Then Reweight: Rethinking Optimization Granularity in On-Policy Distillation*. arXiv:2606.02684, 2026.
- [16] Xu et al. *Speculative Knowledge Distillation: Bridging the Teacher-Student Gap Through Interleaved Sampling*. arXiv:2410.11325, 2024.
- [17] Nie. *Post-Training is About States, Not Tokens: A State Distribution View of SFT, RL, and On-Policy Distillation*. arXiv:2605.22731, 2026.
- [18] Zhang et al. *BREAD: Branched Rollouts from Expert Anchors Bridge SFT & RL for Reasoning*. arXiv:2506.17211, 2025.
- [19] Salimans, Chen. *Learning Montezuma’s Revenge from a Single Demonstration*. arXiv:1812.03381, 2018.
- [20] Resnick et al. *Backplay: “Man muss immer umkehren”*. arXiv:1807.06919, 2018.
- [21] Ecoffet et al. *Go-Explore: a New Approach for Hard-Exploration Problems*. arXiv:1901.10995, 2019; *First return, then explore*. Nature, 2021.
- [22] Choudhury. *Process Reward Models for LLM Agents: Practical Framework and Directions*. arXiv:2502.10325, 2025.
- [23] Lyu et al. *From Correction to Mastery: Reinforced Distillation of Large Language Model Agents*. arXiv:2509.14257, 2025.
- [24] Plyusov et al. *Trust-Region Behavior Blending for On-Policy Distillation*. arXiv:2605.31159, 2026.
- [25] Wang et al. *TCOD: Exploring Temporal Curriculum in On-Policy Distillation for Multi-turn Autonomous Agents*. arXiv:2604.24005, 2026.

- [26] Ross, Gordon, Bagnell. *A Reduction of Imitation Learning to No-Regret Online Learning (DAgger)*. arXiv:1011.0686, 2010.
- [27] Li et al. *Rethinking On-Policy Distillation of Large Language Models: Phenomenology, Mechanism, and Recipe*. arXiv:2604.13016, 2026.
- [28] Ali et al. *Mitigating Copy Bias in In-Context Learning through Neuron Pruning*. arXiv:2410.01288, 2024.
- [29] Wang et al. *Induction Head Toxicity Mechanistically Explains Repetition Curse in Large Language Models*. arXiv:2505.13514, 2025.
- [30] Xu et al. *Adapting the Interface, Not the Model: Runtime Harness Adaptation for Deterministic LLM Agents*. arXiv:2605.22166, 2026.
- [31] Gu. *From Model Scaling to System Scaling: Scaling the Harness in Agentic AI*. arXiv:2605.26112, 2026.
- [32] Li et al. *Preserving Diversity in Supervised Fine-Tuning of Large Language Models*. arXiv:2408.16673, 2024.
- [33] Lin et al. *Same Evidence, Different Answers: Canonical-Context On-Policy Distillation for Multi-Turn Language Models*. arXiv:2605.30251, 2026.
- [34] Ye et al. *On-Policy Context Distillation for Language Models*. arXiv:2602.12275, 2026.
- [35] Liu et al. *MixSD: Mixed Contextual Self-Distillation for Knowledge Injection*. arXiv:2605.16865, 2026.
- [36] Hamman et al. *Few-Shot Knowledge Distillation of LLMs With Counterfactual Explanations*. arXiv:2510.21631, 2025.
- [37] Penalzoa et al. *Privileged Information Distillation for Language Models*. arXiv:2602.04942, 2026.
- [38] Zhao et al. *Self-Distilled Reasoner: On-Policy Self-Distillation for Large Language Models*. arXiv:2601.18734, 2026.
- [39] Lazaridis et al. *EDGE-OPD: Internalizing Privileged Context with Evidence Guided On-Policy Distillation*. arXiv:2605.23493, 2026.
- [40] Zhou et al. *OmniOPD: Logit-Free On-Policy Distillation via Speculative Verification*. arXiv:2606.01476, 2026.
- [41] Jia et al. *Asymmetric On-Policy Distillation: Bridging Exploitation and Imitation at the Token Level*. arXiv:2605.06387, 2026.
- [42] Shen et al. *Anti-Self-Distillation for Reasoning RL via Pointwise Mutual Information*. arXiv:2605.11609, 2026.